Auditing MIFlu and Comparing Influenza Forecasts Across Accuracy Peak Error and Parameter Count

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Single-column review draft oriented towards IEEE JBHI. Current submission rules and page limits remain unverified. This is an evaluation contribution and is not yet submission-ready.

Abstract

We audit an independent MIFlu reproduction and 28 forecasting implementations on a 966-row US national influenza series. The supplied reproduction table gives mean squared error (MSE) 1.4995 versus the published 1.4355, with lower MSE at two horizons and higher MSE at two. Under three benchmark seeds, MIFlu, PatchTST and NLinear achieve 1.5574, 1.6621 and 1.8716. A joint screen using literature-derived timing and intensity tolerances yields 4/16 passing aggregated cells for both NLinear and MIFlu and 0/16 for PatchTST. NLinear has lower mean peak-intensity error but higher mean timing error than MIFlu. Across 11 arms with parameter exports, the MSE–log-parameter fit has slope +0.1171 and R^2 0.0223. An audit identifies an incorrect Holm routine in the supplied analysis: correcting 136 comparisons among 17 models yields no rejections, with NLinear–PatchTST p approximately 0.1172. This does not establish equivalence. Eleven frozen language encoders perform worse than a decision tree in the tested readout configuration. Ablation completion is owner-reported at 60/60 and is not aggregated. The findings are descriptive and limited to one split, retrospective peak detection and adapted implementations. We recommend reporting peak magnitude and timing alongside average error, then selecting on validation-defined costs and budgets before prospective deployment.

I. Introduction

Influenza forecasting supports preparedness, but lower average error does not ensure accurate peak magnitude or timing. We examine a supplied MIFlu reproduction [1] and a broader set of project implementations without treating the joint peak screen as clinical certification.

Contributions. (1) An audited reproduction comparison with a +4.5% mean-MSE gap. (2) A shared-window ranking of 28 implementations. (3) A peak timing–magnitude comparison revealing 4/16, 4/16 and 0/16 aggregate-cell passes for the leading three arms. (4) A parameter and statistical audit identifying low explained variance (R^2 0.0223) and a defective Holm adjustment. The prescriptive addition is a validation-defined decision rule, not a tested new forecasting method.

II. Related Work

Patching and linear baselines are established [3–5]. Time-LLM, CALF and TimeCMA use different alignment/adaptation designs [6–8]. Chronos, TimesFM, Moirai and Lag-Llama pretrain on numerical series and were not evaluated here [9–12]. Earlier LLM-removal work already questions the benefit of language pretraining [13], while a 2026 preprint reports conditional benefits under cross-dataset shifts [14]. The contribution here is the specific influenza protocol audit and multi-metric comparison.

Time-series conformal methods [15–17] motivate future interval evaluation but do not automatically validate intervals for peaks selected retrospectively. WIS [18] evaluates distributions or intervals, while PIE measures point-forecast peak height. These quantities must be separated.

III. Data

The two supplied author-format and paper-exact files each contain 966 rows and seven matched numerical channels, with numeric maximum absolute difference zero. The original downloaded upstream file and its commit are absent. The chronological split is 676/97/193, with train-only scaling and test targets starting no earlier than row 773. Dates in the author-format file are synthetic.

IV. Method

A. Evaluation protocols

We use a chronological split and train-only scaling. Validation selects checkpoints in this implementation. The handoff attributes test-set checkpoint selection and a one-variable loader to upstream code, but a pinned author checkout is absent and the public endpoint could not be opened. Those upstream allegations remain unverified. The notes also show learning-rate differences between code defaults and paper settings, so the comparison must not be described as differing in only two ways.

Reproduction uses seeds 2021–2030. The benchmark uses seeds 42/43/44 and 160/144/144/128 windows at L24/36/48/60. Cropping reproduces the local drop_last rule; a full-window sensitivity analysis is still required. The two MIFlu estimates answer different questions.

B. Metrics

Peak timing error (PTE) is the absolute difference between predicted and observed peak weeks. Peak intensity error (PIE) is the absolute peak-height error divided by the observed peak, expressed as a percentage. Kramer–Shaman [2] verifies the separate one-week and 25% tolerances. We apply both to mean errors in each (model, L, step) cell. This joint screen and its peak detector are project choices, not a clinical acceptance standard. A mean-error pass is not an individual-season success probability.

C. Statistical analysis

The source table contains 136 pairs from 17 models. The other 11 models are absent, so this is not an all-pairs analysis of 28 models. Holm uses a forward cumulative maximum after sorting raw p-values. The source uses a reverse cumulative minimum. At n=12, the two-sided exact floor 2/4096 multiplied by 136 exceeds 0.05, so this family cannot yield a rejection under proper Holm adjustment. Overlapping targets and repeated horizons also make the cells dependent. Raw p-values and unadjusted bootstrap intervals remain exploratory.

D. Decision rule

Decision rule for future validation: specify an MSE budget B, an allowable peak-failure rate q, and decision costs before inspecting a new test period. Among models meeting B and q on validation seasons, minimise expected decision cost, then parameter count as a tie-breaker. For an escalate/wait decision, use costs c_FP for unnecessary escalation and c_FN for a missed surge, with zero cost for correct actions. Escalate when a validated surge probability exceeds c_FP/(c_FP+c_FN). Costs, B and q have not been elicited or validated, so no model is certified for deployment. NLinear is a low-cost candidate for further peak-height validation; MIFlu remains the observed MSE leader.

V. Experiments

The roster comprises 13 conventional models, MIFlu and three other LLM-based implementations, and 11 frozen language encoders. Accuracy uses seven-variable standardised MSE/MAE. Peak errors use de-normalised ILITOTAL. Parameter analysis has 11 models, paired tests 17, and frozen-feature analysis 11; these denominators must not be conflated. We do not present these adapted implementations as verified exact replications of each cited comparator.

VI. Results

A. Reproduction

The reproduction CSV reports mean MSE 1.4995 versus 1.4355, a 4.5% increase. MSE is lower at L24 and L36 and higher at L48 and L60. Mean MAE is 0.8043 versus 0.7452 (+7.9%), but its signs are one lower and three higher; MAE does not share the MSE 2–2 crossover. Protocol and missing upstream provenance limit claims of exact replication.

TABLE I. MIFlu reproduction compared with the paper

Horizon	Our MSE ± SD	Paper MSE	Δ MSE	Our MAE	Paper MAE	Δ MAE
L=24	1.3045 ± 0.0883	1.5420	-15.4%	0.7482	0.7260	+3.1%
L=36	1.3561 ± 0.1603	1.4220	-4.6%	0.7712	0.7790	-1.0%
L=48	1.6543 ± 0.1064	1.4140	+17.0%	0.8381	0.7570	+10.7%
L=60	1.6829 ± 0.2109	1.3640	+23.4%	0.8596	0.7190	+19.6%
4-horizon mean	1.4995	1.4355	+4.5%	0.8043	0.7452	+7.9%

Source: ESSAY_VS_REPRODUCTION_TABLE_V.csv, lines 2–6. SD is recomputed from the code-transcribed ten-seed oracle; its raw run logs are not included. Positive differences mean higher error.

B. Benchmark accuracy

TABLE II. Accuracy of all retained benchmark arms

#	Model	L24	L36	L48	L60	Mean MSE	Mean MAE
1	MIFlu	1.3323	1.2632	1.7774	1.8567	1.5574	0.8199
2	PatchTST	1.4402	1.6154	1.9288	1.6640	1.6621	0.8786
3	NLinear	1.7971	1.7950	1.9574	1.9368	1.8716	0.9183
4	DNN	1.8920	1.7684	2.2603	2.6451	2.1415	1.0057
5	TimeCMA	2.3281	2.5794	2.7104	2.8146	2.6081	1.0676
6	Gradient Boosting	3.5658	3.6867	3.6533	3.7521	3.6645	1.2521
7	KNN	3.7231	3.7623	3.7740	3.8820	3.7854	1.3293
8	Random Forest	3.5486	3.8251	4.0159	4.1710	3.8901	1.2779
9	iTransformer	3.7656	4.0596	4.0438	4.1088	3.9945	1.3476
10	Decision Tree	4.5340	4.1380	4.2186	4.3884	4.3198	1.3798
11	Hunyuan-1.8B	4.5629	4.5877	4.7111	5.0879	4.7374	1.4755
12	Hunyuan-7B	4.3438	4.6624	4.9201	5.1120	4.7596	1.4685
13	DeepSeek-R1-Qwen-1.5B	4.6012	4.6242	4.9245	4.9963	4.7866	1.5002
14	DeepSeek-R1-Llama-8B	4.4337	4.7715	4.8929	5.2019	4.8250	1.4786
15	GRU	4.2895	5.3099	5.1751	4.7736	4.8870	1.5174
16	DeepSeek-R1-32B	4.4401	4.8565	5.0136	5.4386	4.9372	1.4935
17	LSTM	4.4965	5.1359	5.2088	5.1939	5.0088	1.5303
18	DeepSeek-R1-Qwen-14B	4.4976	4.9164	5.1687	5.4838	5.0166	1.5019
19	DeepSeek-R1-Qwen-7B	4.5551	4.9991	5.2417	5.3272	5.0308	1.5133
20	RNN	4.4222	5.2893	5.3424	5.2157	5.0674	1.5634
21	ContextPilot-8B	4.6094	5.1318	5.2895	5.3065	5.0843	1.5255
22	Hunyuan-4B	4.7383	5.0535	5.2876	5.4474	5.1317	1.5450
23	ContextPilot-14B	4.7674	5.1566	5.2168	5.6388	5.1949	1.5407
24	Qwen2.5-1.5B	4.7840	5.0587	5.4546	5.5224	5.2049	1.5725
25	Qwen-Time	4.7190	5.1325	5.6894	5.4818	5.2557	1.5794
26	SVR	5.2248	5.3581	5.3781	5.3940	5.3388	1.4986
27	CALF	5.3624	5.5113	5.6442	5.7962	5.5785	1.6190
28	CNN	5.5021	5.3098	5.2801	6.3698	5.6155	1.6720

Source: BENCH_MSE_BY_MODEL_L.csv, keyed by model and L. Each row averages three seeds on 160/144/144/128 windows. DLinear is excluded by project choice; that exclusion needs a documented sensitivity analysis. Names denote project implementations, whose equivalence to published methods is not established by this ranking.

F0 conclusion. MIFlu has the lowest observed MSE (1.5574), but MSE alone does not establish public-health utility. The top-three spread is 20.2% relative to MIFlu, not less than 20%.

C. Peak timing and magnitude

TABLE III. Peak timing and intensity error

Model	PTE (wk)	PIE (%)	Joint passes	s0 PTE	s0 PIE (%)
MIFlu	1.128	26.846	4/16	0.347	18.884
PatchTST	1.240	32.755	0/16	0.681	30.553
NLinear	1.677	9.477	4/16	0.208	3.250

Means are across 16 (L, step) cells. Passes apply PTE ≤ 1 week and PIE ≤ 25% to each cell's aggregated errors, not to each individual season. PIE is point-forecast magnitude error, not probabilistic calibration.

TABLE IV. Joint peak screen for each horizon and lead

Model	L	s0	s1	s2	s3
NLinear	24	PASS 0.111 / 3.949	fail 1.333 / 8.411	fail 2.333 / 11.943	fail 2.667 / 18.739
NLinear	36	PASS 0.333 / 4.181	fail 1.333 / 7.222	fail 2.111 / 15.313	fail 2.667 / 18.145
NLinear	48	PASS 0.222 / 2.419	fail 1.222 / 4.727	fail 2.222 / 9.979	fail 2.778 / 16.655
NLinear	60	PASS 0.167 / 2.450	fail 1.444 / 4.542	fail 2.556 / 9.952	fail 3.333 / 13.000
MIFlu	24	PASS 0.000 / 18.307	fail 0.889 / 26.686	fail 1.222 / 30.472	fail 1.333 / 31.638
MIFlu	36	PASS 0.000 / 18.695	fail 0.778 / 26.333	fail 1.667 / 31.514	fail 1.333 / 32.427
MIFlu	48	fail 1.222 / 18.004	PASS 0.667 / 24.795	fail 1.222 / 30.133	fail 1.333 / 33.236
MIFlu	60	PASS 0.167 / 20.530	fail 1.333 / 25.799	fail 2.222 / 29.790	fail 2.667 / 31.177
PatchTST	24	fail 0.667 / 30.580	fail 1.000 / 34.044	fail 1.222 / 33.572	fail 1.222 / 30.722
PatchTST	36	fail 1.000 / 26.235	fail 1.667 / 32.247	fail 1.333 / 31.245	fail 1.333 / 33.854
PatchTST	48	fail 0.556 / 30.360	fail 1.333 / 30.878	fail 0.889 / 35.158	fail 0.889 / 35.002
PatchTST	60	fail 0.500 / 35.035	fail 1.667 / 34.342	fail 2.222 / 35.674	fail 2.333 / 35.133

Each entry gives PTE in weeks / PIE in percent. s0–s3 are the first to fourth forecast weeks. The detector uses prominence 0.15 × range, distance 15, and ±8-week matching. These detector settings and joint aggregation are project choices. Source: BENCH_PTE_PIE_BY_STEP.csv.

F1 conclusion. NLinear and MIFlu each pass 4/16 aggregated cells, while PatchTST passes 0/16. NLinear has lower mean PIE (9.477% versus MIFlu 26.846%) but higher mean PTE (1.677 versus 1.128 weeks). This is a timing–magnitude trade-off, not dominance. The ordering is inverted at both ends among the three leaders: MSE orders MIFlu, PatchTST, NLinear; PIE orders NLinear, MIFlu, PatchTST.

MSE correlates with PIE (ρ=+0.736, $p<0.0001$) and inversely with passes (ρ=−0.429, p=0.023). Both signs are consistent with generally better error accompanying better peak results. These associations coexist with NLinear's lower PIE despite its third-place MSE. They do not imply independence or a complete rank reversal.

D. Parameters

TABLE V. Parameter analysis and rank associations

Quantity	Value
Regression slope / R^2	+0.1171 / 0.0223
Models with parameter exports	11
Subset Pareto set	NLinear 2,520; PatchTST 531,878
Parameter ratio	211.1×
Descriptive MSE reduction	11.2%
ρ(MSE, PIE)	+0.736 (p < 0.0001)
ρ(MSE, passes)	-0.429 (p = 0.023)

The Pareto analysis includes only 11 arms with parameter exports. MIFlu and the frozen encoders are not in that subset. Missing parameter counts are not zero. Parameter count is not a measurement of latency, memory or energy. Correlations are descriptive across models on one test set.

F2 conclusion. The fitted slope is +0.1171 and R^2 is 0.0223 across the 11 arms with exported parameter counts. Size alone explains little of their observed MSE variation. This subset result does not prove that capacity is irrelevant.

E. Statistical correction

TABLE VI. Paired comparisons with corrected Holm adjustment

Model A	Model B	Δ MSE	p raw	p Holm	Effect*
NLinear	PatchTST	+0.2095	0.005859	0.1172	1.15
Gradient Boosting	RNN	-1.4029	0.000488	0.0664	4.86
KNN	NLinear	+1.9138	0.000488	0.0664	22.29
KNN	MIFlu	+2.2280	0.000488	0.0664	9.89
KNN	LSTM	-1.2234	0.000488	0.0664	5.28
iTransformer	TimeCMA	+1.3863	0.000488	0.0664	7.02

Model A	Model B	Δ MSE	p raw	p Holm	Effect*
iTransformer	SVR	-1.3443	0.000488	0.0664	10.61

*136 pairs cover 17 models, not all 28. n = 12 matched horizon × seed cells. Forward cumulative-max Holm yields 0/136 rejections. The shown p values recover the exact 1/4096 enumeration grid from six-decimal exports; recomputing directly from rounded exports gives NLinear–PatchTST 0.117180 instead of 0.1171875. *The source effect uses pooled model SD, not paired-difference SD. Cell dependence limits inference.*

F3 conclusion. Proper Holm gives 0/136 rejections (128/136 before adjustment), with a minimum adjusted p of about 0.0664 and NLinear–PatchTST p ≈ 0.1172. PatchTST still has 11.2% lower observed MSE with 211× the parameters and lower MSE in 10/12 cells. Non-rejection does not establish equivalence or show that the gain is noise.

F. Frozen language encoders

TABLE VII. Frozen encoders with shared linear readout

Encoder	MSE	vs MIFlu	PTE	PIE (%)	Passes
Hunyuan-1.8B	4.7374	3.04×	4.406	71.428	0/16
Hunyuan-7B	4.7596	3.06×	5.309	75.399	0/16
DeepSeek-R1-Qwen-1.5B	4.7866	3.07×	5.792	78.061	0/16
DeepSeek-R1-Llama-8B	4.8250	3.10×	4.306	77.121	0/16
DeepSeek-R1-32B	4.9372	3.17×	3.889	77.730	0/16
DeepSeek-R1-Qwen-14B	5.0166	3.22×	4.149	76.350	0/16
DeepSeek-R1-Qwen-7B	5.0308	3.23×	4.458	76.373	0/16
ContextPilot-8B	5.0843	3.26×	4.962	78.918	0/16
Hunyuan-4B	5.1317	3.30×	3.003	78.840	0/16
ContextPilot-14B	5.1949	3.34×	4.965	77.764	0/16
Qwen2.5-1.5B	5.2049	3.34×	5.333	74.650	0/16

11 retained encoders × 4 horizons × 3 seeds = 132 cells. The historical plan included two additional embedding-only arms, giving 156 planned cells; those 24 omitted cells are outside the 28-model roster. The result tests only this frozen-feature pipeline.

F4 conclusion. The 11 tested frozen encoders score 4.7374–5.2049, or 3.04–3.34× MIFlu's MSE. All exceed the decision tree's 4.3198 and pass 0/16 cells. This is a negative result for this feature/readout configuration; prompting, fine-tuning and numerical foundation models remain untested.

G. Ablation design

TABLE VIII. Ablation design and snapshot status

Variant	Flag	Cells	Status
Reference K=6	default	12	Already in benchmark
no_lora	--no-lora	12	DONE (not aggregated)
K=2	--num-layers 2	12	DONE (not aggregated)
K=4	--num-layers 4	12	DONE (not aggregated)
K=8	--num-layers 8	12	DONE (not aggregated)
K=12	--num-layers 12	12	DONE (not aggregated)

Owner-reported status at 3 November 2026: 60/60 additional cells, job 722968. Burgundy queue and per-cell manifests show 60/60 DONE; the per-variant aggregate trend has not been recomputed. The baseline 12 cells are not part of the additional 60. No trend is inferred.

F5 provisional conclusion. In miflu_model.py's author_faithful path, text follows time tokens behind causal attention and the head reads the time tokens. Static inspection therefore predicts no text influence during deterministic evaluation. The supplied tests_archive/test_miflu_ablations.py explicitly asserts both text invariance and full/no-text equality with matched CPU initialisation. It was inspected statically, not executed here. Other fusion paths and training randomness require separate checks.

VII. Threats to Validity

One national series and one chronological split do not establish regional or prospective utility. Peak matching can use future observations to define events and is a retrospective evaluation tool. Synthetic date labels require a verified CDC calendar mapping before season-specific claims. Parameter exports omit MIFlu and encoders. The 60/60 ablation status is

owner-reported, not a live cluster check. Modern numerical foundation baselines, regional results, rolling origins and validated uncertainty are missing.

Source p-values are retained but their complete raw paired inputs were not independently rescored. Corrected Holm values recover the enumeration grid and retain provenance to the rounded export. The reported source effect uses pooled model SD rather than paired-difference SD. A seed ensemble with an arbitrary variance floor is not a validated predictive distribution. No prospective utility, equivalence margin or calibrated surge probability has been established.

VIII. Conclusion

MIFlu leads the observed MSE ranking. NLinear offers lower peak-height error at low parameter count but has worse mean peak timing than MIFlu. The joint screen therefore exposes trade-offs rather than a universal winner. Corrected significance analysis supports no familywise rejection at the present resolution. Further regional, rolling-origin and foundation-model controls are needed before submission or deployment.

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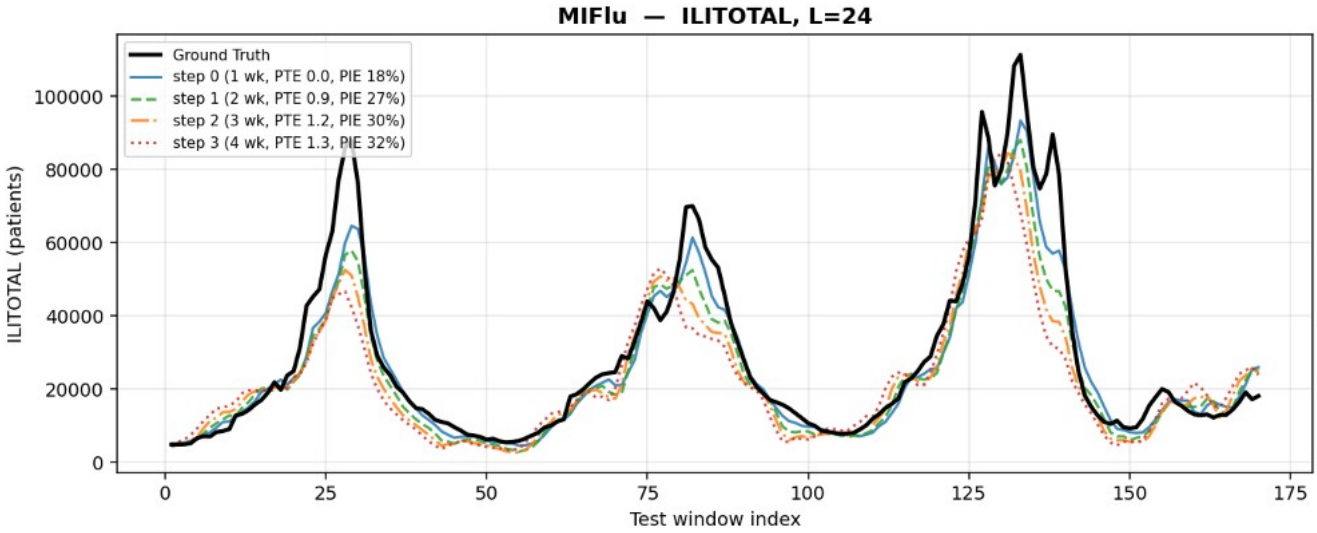
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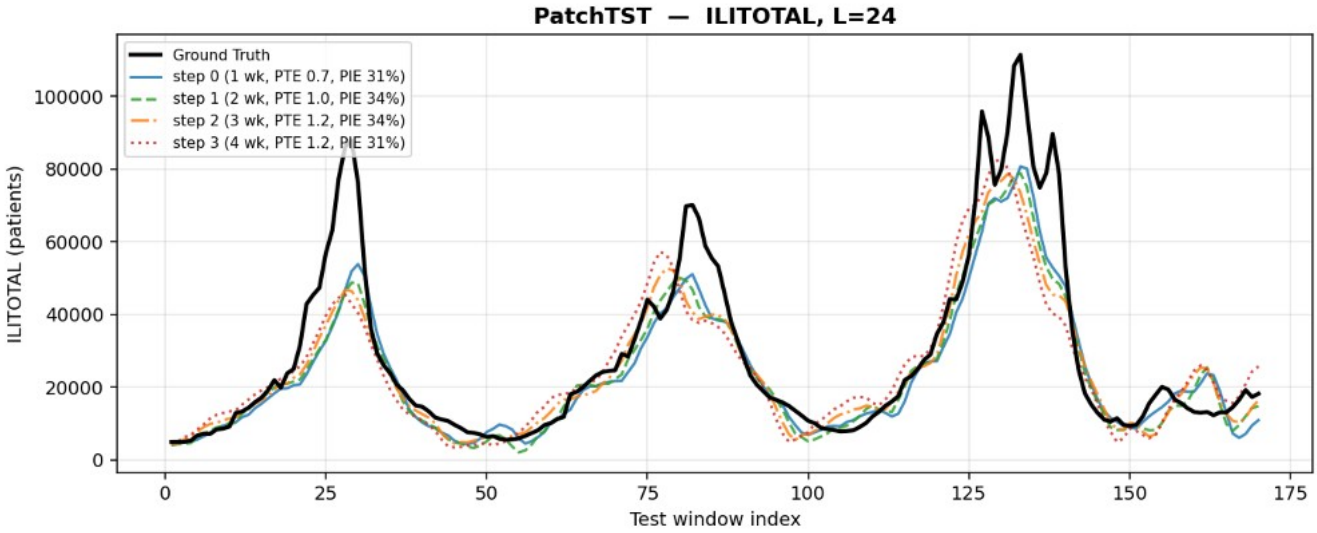
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Appendix A Forecast panels

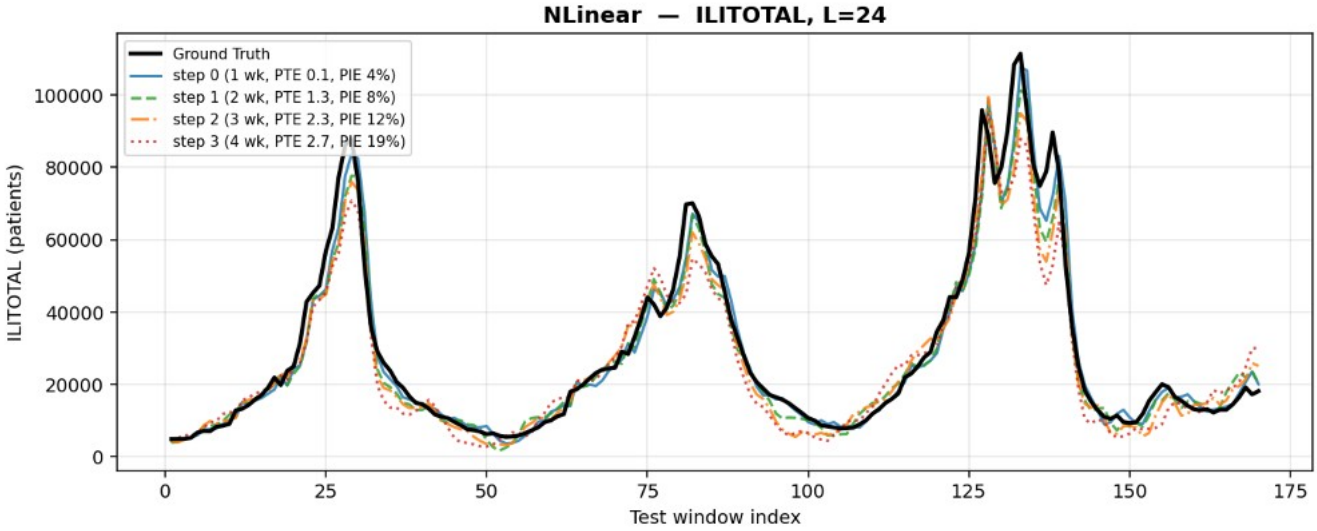
Illustrative panels from the supplied snapshot. The x-axis is the file's test-week index, not independently verified CDC epidemiological dates. These figures do not establish causality or prospective performance.



miflu, L=24. ILITOTAL in patient counts against the file's test-week index. Black is the observed series; the other curves use fixed forecast steps s0–s3. Scored windows: 160. Imported illustrative figure; raw figure inputs were not independently regenerated in this audit.



patchtst, L=24. ILITOTAL in patient counts against the file's test-week index. Black is the observed series; the other curves use fixed forecast steps s0–s3. Scored windows: 160. Imported illustrative figure; raw figure inputs were not independently regenerated in this audit.



nlinear, L=24. ILITOTAL in patient counts against the file's test-week index. Black is the observed series; the other curves use fixed forecast steps s0–s3. Scored windows: 160. Imported illustrative figure; raw figure inputs were not independently regenerated in this audit.